

Here, ECHO OFF command does not allow the command prompt to appear on the screen. When @ is prefixed, echoing of a line on the screen is prevented. Therefore @ECHO OFF command prevents echoing of batch file contents on the screen.

Path is an environment variable specifying a set of directories where executable programs are located. It is a series of directories separated by semicolons (;). Microsoft Windows looks for programs in Path directories in order, from left to right, and executes them. In this batch file, the path variable is set to JDK bin directory, where executable commands javac (to compile the program) and java (to run the program) are located.

javac command is used to compile the program. The program is distributed among two files, inrush_4a.java and LineChart_use.java, both of which are compiled together using the command:

```
javac inrush_4a.java LineChart_use.java
```

In order to run the compiled java inrush_4a program, double-click inrush4a.bat file or right-click the icon of inrush.bat file, and select 'Open'. A window will appear on the screen with the following options:

1. Enter Transformer Data
2. Display Inrush Current Graph
3. Calculate-Effect of Switching Angle

4. Harmonic Analysis

This is the main window (Fig. 4).

The desired option can be selected by clicking.

Enter Transformer Data. First of all, enter data of the transformer for which inrush current analysis is to be done. Click Enter Transformer Data in the main window. Now, another window is displayed on the screen, wherein you need to fill in transformer details under following 13 fields:

1. Transformer ID No. < Auto Generated by Program > {not to be entered}

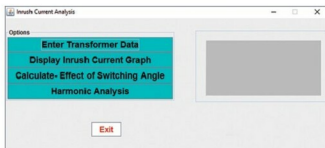


Fig. 4. Main program output window

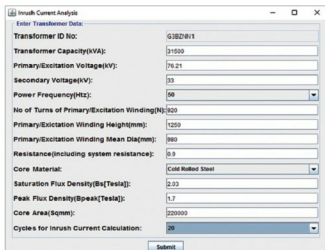


Fig. 5. Inrush current calculation

2. **Transformer Capacity.** Here the transformer capacity in kVA is to be entered

3. **Primary/Excitation Voltage.** Here enter voltage of the primary/excitation side, that is, voltage of the side that is connected to the source, in kV. In case of a 3-phase transformer, enter the phase value of the voltage. For example, if it is a 132kV/33kV Star/Delta (Y/D) transformer that is excited from the primary side, the voltage value will be $132/\sqrt{3} = 76.21$ kV.

4. **Secondary Voltage.** Here enter voltage of the secondary side in kV. In case of a 3-phase transformer, enter the phase value of the voltage. For example, if it is a 132kV/33kV Star/Delta (Y/D) transformer, secondary-side voltage will be 33kV.

5. **Power Frequency.** Here select power frequency of the supply from the drop-down menu. It may be 50Hz or 60Hz. For example, it is 50Hz for India and 60Hz for USA.

6. **Number of Turns of Primary/Excitation Winding.** Here enter the number of turns of the winding that

is connected to supply side.

7. **Primary/Excitation Winding Height.** Here enter axial height of the primary/excitation winding in mm.

8. **Primary/Excitation Winding Mean Dia.** Here enter mean diameter of the excitation winding in mm.

9. **Resistance.** Here enter resistance of the primary/excitation winding including resistance of the system, that is, resistance of the line between source and transformer, in ohms.

10. **Core Material.** Usually, the transformer core is made of grain-oriented steel. Select core material of the transformer from the drop-down list as 'Hot Rolled Steel' or 'Cold Rolled Steel'.

11. **Saturation Flux Density.** Here enter saturation flux density (Bs) of the core material in Tesla.

12. **Peak Flux Density.** Here enter peak value of designed steady-state flux density (Bpeak) of the core material in Tesla.

13. **Cycles for Inrush Current Calculation.** From drop-down menu, select the number of cycles for which inrush current is to be calculated and plotted. The number of cycles may be selected between 5 and 50. This window is shown in Fig. 5.

14. **Core area.** Here the core area of transformer is entered in sqmm.

Once you enter transformer data, click Submit. In case the entered data is correct, the data entry screen disappears and the main window appears on the screen.

The program automatically checks data entry. Data is accepted only if none of the fields is left blank, and numeric data is entered wherever required. For example, if non-numeric data is entered in the field for Transformer Capacity, it will not be accepted.

Display Inrush Current Graph. If this option is selected, the graph showing peak value of inrush current in each cycle is plotted. The number of cycles is given on x-axis, while